

FINAL SEMESTER PROJECT

MID DEFENSE REPORT FORMAT

TITLE PAGE

CERTIFICATION / RECOMMENDATION PAGE

This is to certify that this Mid Defense Report entitled "**Project Title**" submitted by **Student Name** in partial fulfillment of the requirements for the degree of **Bachelor of Engineering in Computer Engineering** is a bonafide work carried out under my supervision.

Supervisor Signature: _____

Name:

Date:

ACKNOWLEDGEMENT

The student expresses sincere gratitude to the supervisor, department, campus administration, and all individuals who directly or indirectly supported the completion of this mid defense work.

ABSTRACT

The abstract provides a concise yet comprehensive summary of the final semester project with a specific focus on the mid defense stage. It should briefly explain the background of the problem, the motivation behind selecting the topic, and the main objectives of the project. This section must clearly highlight the methodology adopted so far, including data collection, system/model design, and initial implementation work completed up to the mid defense. It should also mention the tools, technologies, or algorithms used and outline the expected outcomes by the final defense. The abstract should typically be written in a single paragraph of around 200–300 words and must give the reader a clear overall picture of the project status.

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List of Figures (if any, must be auto generated)

List of Tables (if any, must be auto generated)

Abbreviations

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

This subsection introduces the general domain of the project and provides contextual background. It explains the real world scenario or technical area in which the problem exists and why it is important. The discussion should gradually move from a broad overview of the field to the specific area addressed by the project.

1.2 Project Overview

This subsection gives a brief overview of the proposed system or research work. It explains what the system aims to do, who the target users are (if applicable), and how the system or model is expected to solve the identified problem. A high-level description of the approach adopted in the project should be included without going into technical depth.

1.3 Project Scope and Application

This subsection defines the boundaries and coverage of the project. It explains what aspects are included within the project scope and what aspects are intentionally excluded. The scope clarifies the limitations of the system in terms of functionality, data, users, or environment. This section also describes the practical applications of the project, such as academic use, industry relevance, decision support, automation, or real-world problem-solving scenarios where the developed system or model can be applied.

1.4 Feasibility Study (Technical, Economical, Operational)

This subsection evaluates the practicality of implementing the proposed project.

a) Technical Feasibility:

Assesses whether the required hardware, software, tools, technologies, and technical skills are available to successfully develop the project. It justifies the selection of programming languages, frameworks, and platforms.

b) Economical Feasibility:

Analyzes the cost-effectiveness of the project. It considers development cost, software licensing (if any), hardware requirements, and whether the project can be completed within reasonable financial constraints.

c) Operational Feasibility:

Examines how well the proposed system will operate in a real environment. It focuses on ease of use, user acceptance, system reliability, and whether the solution effectively meets user and organizational needs.

CHAPTER 2: PROBLEM STATEMENT

This chapter clearly and precisely defines the problem that the project intends to address. The problem statement should highlight existing limitations, challenges, or gaps in current systems or research. It must justify the need for the proposed project by explaining why the problem is significant and what difficulties users, organizations, or researchers currently face. The problem should be stated in a focused and unambiguous manner.

CHAPTER 3: OBJECTIVES OF THE STUDY

3.1 General Objective

The general objective states the overall goal of the project. It summarizes what the project aims to achieve upon successful completion.

3.2 Specific Objectives

Specific objectives break down the general objective into smaller, achievable tasks. These objectives should be clear, measurable, and aligned with the project scope. They often include activities such as designing the system, implementing modules, evaluating performance, and analyzing results.

CHAPTER 4: LITERATURE REVIEW

This chapter presents a review of existing studies, systems, research papers, and technologies related to the project topic. It explains how similar problems have been addressed in the past and what methodologies or tools were used. The literature review should also identify limitations or research gaps in existing works, which directly justify the need for the proposed project. Proper citation of sources is essential in this section.

CHAPTER 5: METHODOLOGY

5.1 Overall Methodology

This subsection explains the step-by-step approach followed to carry out the project. It may include requirement analysis, system design, data collection, preprocessing, implementation, testing, and evaluation. A flowchart or methodology diagram can be referenced here.

5.2 Proposed Model / Algorithm

This subsection describes the proposed system model, algorithm, or framework in detail. It explains how the proposed approach works and why it is suitable for solving the identified problem. At the mid defense stage, emphasis should be given to the design and partially implemented components.

CHAPTER 6: SYSTEM DESIGN / MODEL ARCHITECTURE

This chapter explains the architectural design of the system or model. It may include block diagrams, UML diagrams, model architecture diagrams, data flow diagrams, or workflow representations. Each component of the system should be briefly explained along with its role. This section demonstrates the technical planning and design clarity achieved before full implementation.

CHAPTER 7: DATASET DESCRIPTION

This chapter describes the dataset used or to be used in the project. It includes the data source, size, attributes/features, and data format. If preprocessing steps such as cleaning, normalization, or transformation have been performed, they should be explained here. For projects without datasets, this section may describe input data or system parameters.

CHAPTER 8: TOOLS AND TECHNOLOGIES USED

This chapter lists and explains the tools, technologies, and platforms used in the project. It may include programming languages, frameworks, libraries, databases, development tools, and hardware/software requirements. Justification for selecting specific tools or technologies should also be provided.

CHAPTER 9: IMPLEMENTATION AND TESTING

9.1 Implementation Details

This section describes how the proposed system or model has been implemented up to the mid defense stage. It explains the major modules or components developed, algorithms implemented, and workflows followed. Any challenges faced during implementation and the solutions adopted should be briefly discussed.

9.2 Testing Strategy

This subsection explains the testing approach adopted to verify the correctness and reliability of the system. It may include unit testing, module testing, integration testing, and system testing as applicable. For mid defense, testing may be partial, focusing on implemented components.

9.3 Test Cases (If Applicable)

This subsection presents sample test cases with inputs, expected outputs, and observed results. It demonstrates that the system has been logically validated, even if full testing is planned for the final defense.

CHAPTER 10: RESULT ANALYSIS AND EVALUATION

This chapter analyzes the preliminary results obtained from the implemented parts of the project. It evaluates system performance based on relevant criteria such as accuracy, efficiency, response time, or correctness, depending on the nature of the project. Graphs, tables, or screenshots may be included to support the analysis. At the mid defense stage, this evaluation is initial and indicative, with detailed analysis planned for the final defense.

CHAPTER 11: WORK COMPLETED TILL DATE (MID DEFENSE)

This is a key chapter for mid defense. It clearly outlines all the tasks and activities completed up to this stage. Typical contents include completion of literature review, problem analysis, dataset collection, system or model design, partial implementation, and preliminary results if available.

CHAPTER 12: REMAINING WORK PLAN (FINAL YEAR)

This chapter explains the tasks that will be carried out after the mid defense until the final defense. It includes detailed planning of remaining implementation, testing, evaluation, result analysis, documentation, and final presentation. A clear and realistic work plan reflects proper time management and project feasibility.

CHAPTER 13: CONCLUSION

This chapter summarizes the overall work completed up to the mid defense. It briefly revisits the problem statement, objectives, methodology, and progress achieved so far. The conclusion should emphasize the importance of the project and indicate confidence in completing the remaining work successfully by the final defense.

REFERENCES

Use IEEE referencing style

Report Format Standards:

- Page Number
 - The pages from certificate page to the list of
- tables/figures/abbreviations/approvals should be numbered in roman starting from i.
 - The pages from chapter 1 onwards should be numbered in numeric starting from 1. The page number should be inserted at bottom, aligned center.
- B. Page Size and Margin
 - The paper size must be a page size corresponding to A4.
 - The margins must be set as
- Top = 1 in (2.54 cm)
- Bottom = 1 in (2.54 cm)
- Left = 1.25 in (3.17 cm)
- Right = 1 in (2.54 cm)
- Paragraph Style
 - All paragraphs must be justified and have spacing of 1.5.
- Text Font of Document
 - The contents in the document should be in Times New Roman font • The font size in the paragraphs of document should be 12
- Section Headings
 - Font size for the headings should be 16 for chapter headings, 14 for section headings, 12 for sub-section headings.
 - All the headings should be bold faced.
- Figures and Tables
 - Position of figures and tables should be aligned center.
 - The figure caption should be centered below the figure and table captions should be centered above the table.
 - All the captions should be of Italic with 9 font size.

Notes:

- Start new chapter from new page.
- Save document in pdf format before print.
- **Special NOTE: While printing the report you should be sure that you have chosen A4 rather than letter format.**